Semester III													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24INT152T	Rural Internship	0	0	0	0	1				50	50	100
24MA201T	Mathematics-III	3	1	0	4	4	25	25	50	--	--	100
24PE201T	Introduction to Artificial Intelligence	3	0	0	3	3	25	25	50	--	--	100
25PE201T	Fundamentals of Fluid Mechanics for Petroleum Engineers	3	0	0	3	3	25	25	50	--	--	100
24PE203T	Thermodynamics of Petroleum Reservoir Fluids	2	1	0	3	3	25	25	50	--	--	100
24PE204T	Geomechanics and Strength of Materials	2	1	0	3	3	25	25	50	--	--	100
24PE204P	Geomechanics and Strength of Materials Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PE205T	Petroleum Exploration	3	0	0	3	3	25	25	50	--	--	100
24PE205P	Petroleum Exploration Laboratory	0	0	2	2	1	--	--	--	50	50	100
	Total	**16**	**3**	**4**	**23**	**22**						

Semester IV													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
	Open Elective 1 (From Other School)	3	0	0	3	3	25	25	50	--	--	100
24PE206T	Industry 4.0	2	0	0	2	2	25	25	50	--	--	100
24PE206P	Industrial 4.0 Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PE207T	Drilling Engineering	3	1	0	4	4	25	25	50	--	--	100
24PE207P	Drilling Engineering Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PE208T	Reservoir Engineering	3	1	0	4	4	25	25	50	--	--	100
24PE209T	Well Log and Formation Evaluation	3	1	0	4	4	25	25	50	--	--	100
24PE202T	Heat and Mass Transfer	2	1	0	3	3	25	25	50	--	--	100
24PE210T	Earth Science and Hydrocarbon Exploration Field Work	0	0	6	0	3	--	--	--	50	50	100
	Total	**16**	**4**	**10**	**24**	**25**						

24PE206T					Industry 4.0					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand the core technologies powering Industry 4.0.
- To analyze applications of Industry 4.0 technologies to improve the efficiency, safety, and sustainability of oil and gas operations.
- To develop strategies for Industry 4.0 implementation in oil and gas companies.

Unit I: Introduction and Basic Concepts of Industry 4.0 **Hours: 08**

Introduction to Industry 4.0, Definition, General framework, Application areas, Introduction to the Evolution of Industrial revolutions, Key features, Need and benefits, **Introduction to Industry 4.0 core technologies:** Big data, Advanced Robotics, Simulation, Integration, Internet of Things (IoT), Artificial Intelligence (AI), Cybersecurity, Cloud computing, Additive manufacturing and Augmented Reality. Overview of machine learning for Industry 4.0. Transformation of industrial processes through the integration of modern technologies such as sensing and actuation, communication, and computational processing.

Unit II: Industry 4.0 Technologies in the Oilfield Landscape **Hours: 06**

Digital Oilfield Transformation, Digital Oilfield concept (DOF), **Sensors and the Internet of Things (IoT):** Types of sensors used in oilfields (temperature, pressure, flow, seismic), IoT network architectures, Connected Oilfield: Integration of sensors, actuators, and communication networks across oilfield operations.

Unit III: Intelligent Operations and Automation **Hours: 06**

Drilling Optimisation, Smart Reservoir management, Automation of Well operations, Greenfield and brownfield automation, Process Control and Safety

Unit IV: Digital Twins for Enhanced Operations **Hours: 06**

Concept of digital twins, Simulation and prediction of performance, optimization of maintenance schedules, risk mitigation, Integration of sensor data, physics-based models for real-time updates on the digital twin

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the core elements and basic technologies for Industry 4.0
CO2 - Apply the different computational techniques and algorithms for realizing Industry 4.0
CO3 - Transform the traditional business approach by integrating data and intelligence
CO4 - Develop the traditional industries with intelligent and automated machines
CO5 - Utilize data and intelligence for the development of a Smart World
CO6 - EUnderstand the concept, significance and means to achieve sustainable development

TEXT / REFERENCE BOOKS

1. Bali, V., et al. (2021). *Industry 4.0, AI, and data science: Research trends and challenges*. CRC Press.
2. Bangert, P. (2019). *Machine learning and data science in the oil and gas industry: Best practices, technical innovations, and case studies*. Gulf Professional Publishing.
3. Hemmati-Sarapardeh, A., et al. (2020). *Applications of artificial intelligence techniques in the petroleum industry*. Gulf Professional Publishing.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE206P					Industrial 4.0 Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To provide a strong foundation in Industry 4.0 concepts as they apply to the petroleum industry.
- To develop proficiency in utilizing Industry 4.0 technologies for process optimization, predictive maintenance, and reservoir management.
- To encourage critical thinking and problem-solving skills in the context of digital transformation in the petroleum sector.

LIST OF EXPERIMENTS

1. To analyze Data Analytics for Reservoir Management – part A
2. To analyze Data Analytics for Reservoir Management – part B
3. To develop a simplified digital twin model for production optimisation – part A
4. To develop a simplified digital twin model for production optimisation – part B
5. To explore historical pipeline sensor data and use Machine Learning (ML) for Pipeline Integrity Prediction
6. To design a basic Internet of Things (IoT) system for well-monitoring
7. To apply 3D printing in creating customized downhole tools – part A
8. To apply 3D printing in creating customized downhole tools – part B
9. To explore the Augmented Reality (AR) in oil and gas operations for Remote Assistance – part A
10. To explore the Augmented Reality (AR) in oil and gas operations for Remote Assistance – part B
11. To research Oil and Gas robots for well inspection and maintenance tasks in hazardous environments
12. To use big data analytics in processing and interpreting large seismic datasets – part A
13. To use big data analytics in processing and interpreting large seismic datasets – part B

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the concept of Industry 4.0 and its significance
CO2 - Understand the resource requirements for the implementation of Industry 4.0
CO3 - Learn the Simulation Packages for Industry 4.0
CO4 - Explore the concept of Smart Infrastructure through simulation studies
CO5 - Inspect embedded platform applications for Industry 4.0
CO6 - Synthesise the solution for the given Industry 4.0 related problem

TEXT / REFERENCE BOOKS

1. Bali, V., et al. (2021). *Industry 4.0, AI, and data science: Research trends and challenges*. CRC Press.
2. Bangert, P. (2019). *Machine learning and data science in the oil and gas industry: Best practices, technical innovations, and case studies*. Gulf Professional Publishing.
3. Hemmati-Sarapardeh, A., et al. (2020). *Applications of artificial intelligence techniques in the petroleum industry*. Gulf Professional Publishing.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**

24PE207T					Drilling Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	-	-	100

COURSE OBJECTIVES

- Demonstrate the equipment and practices of oil well drilling
- Illustrate the Casing practices and Cementing techniques
- Imbibe skills to prepare a Geo-Technical order
- Develop skills to address problems in drilling oil wells

Unit I: Drilling Rig and Drilling Fluids **Hours: 16**

Evolution of Well Drilling Processes; Drilling Operations; Drilling Rig Components/Systems: Power, Derrick & Sub-structure, Hoisting System, Rotary systems, Circulation & Mud treatment, Rig control system; Drilling Fluids: Basics, Functions, Properties and Nature; Classification of drilling fluids viz. Pneumatic drilling fluid, Water-based drilling fluids, Oil-based drilling fluids; Latest advances and emerging trends in drilling fluids viz. use of NDDF, Inhibitive, and Non-inhibitive; Balanced/Underbalanced drilling; Rheological models of drilling fluids: Bingham Plastic Model, Power law model, Herschel-Buckley model, Mud weight Calculations; Well Planning and Geo-Technical Order (GTO); Pore Pressure prediction, Fracture pressure, Overburden Pressure, and Abnormal pressure.

Unit II: Drill String, Casing, and Bit Design **Hours: 15**

Drill String: Components and functions; Drill String design; Drill String fatigue failure; Drill Bits: Types, Performance and Criteria for design, Bit Nozzle size selection, Rig Hydraulics; Bit failure; Casing: Functions, Configuration, Types, Properties, Casing setting depth and hole sizes, Casing handling practices; Casing design criteria; Casing and Liner design.

Unit III: Cementing Techniques **Hours: 11**

Introduction to Cementing, Cement and Cement Slurry: Objectives of cementing, Classification of cement, Slurry design, Slurry additives, Factors influencing cement slurry design; Cementing equipment; Cementing Methods: Primary cementing, Stage cementing, Liner cementing, Plug Cementing, Squeeze Cementing techniques in practice; Deep well cementing; Characteristics of good quality cementation using CBL/VDL. Cementing calculation

Unit IV: Drilling Problems and Rig Site HSE **Hours: 10**

Pipe sticking and Calculations, Pipe failure, Lost circulation, Hole Deviation, Sloughing shale, Formation damage, Borehole Instability; Fishing and coring operations; Hole Cleaning; General equipment and personnel safety; Environmental Impact of Drilling fluid; Drilling waste management: Classification of drilling waste, Approaches of Drilling waste minimization.

Max <52Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recognize the drilling practices on drill site.
CO2 - Express information with increased technical clarity.
CO3 - Relate changes with changes in drilling parameters
CO4 - Investigate drilling bottlenecks during drilling
CO5 - Plan a drilling program
CO6 - Recommend optimum conditions to drill a well

TEXT / REFERENCE BOOKS

1. Rabia, H. (1986). *Oil well drilling engineering: Principles and practices*. Graham & Trotman.
2. Bourgoyne, A. T., Jr., Chenevert, M. E., Millheim, K. K., & Young, F. S., Jr. (1991). *Applied drilling engineering*. Society of Petroleum Engineers.
3. Azar, J. J., & Samuel, G. R. (2007). *Drilling engineering*. PennWell Books.
4. International Association of Drilling Contractors. (2000). *IADC drilling manual*. International Association of Drilling Contractors.
5. Adams, N. J. (1985). *Drilling engineering: A complete well planning approach*. PennWell Books.
6. Gatlin, C. (1960). *Petroleum engineering: Drilling and well completion* (1st ed.). Prentice Hall.
7. Darley, H. C. H., & Gray, G. R. (1988). *Composition and properties of drilling and completion fluids* (5th ed.). Gulf Publishing Company.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE207P					Drilling Engineering Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To provide the concept of measuring different properties of drilling fluid and cement slurry
- Explain the importance and limitations of all parameters of drilling fluid and cement slurry
- Provide the experimental procedure and data analysis of performed experiments
- To evaluate the drilling fluid and cement slurry properties as per standards

LIST OF EXPERIMENTS

1. Preparation of WBM and determination of pH of drilling fluid.
2. Determination of the mud weight and viscosity of the given mud.
3. Determination of the Rheology of the drilling fluid.
4. Determination of the Fluid loss, Mud cake thickness, and Sand content of a WBM.
5. Determination of the Total hardness in drilling fluid by titration.
6. Determination of the alkalinity in Drilling fluid and in filtrate of drilling fluid by titration.
7. Determination of the cation exchange capacity using field procedure.
8. Preparation of OBM and measuring the gel or shear strength of OBM.
9. Determination of the volumes of Water, Oil and Solids in OBM
10. Preparation of homogeneous cement slurry and determination of slurry density.
11. Demonstration of Atmospheric and HPHT Consistometer.
12. Estimation of the volume of cement filtrate loss to the formation.
13. Determination of the rheological properties and the behavior of cement slurries.
14. Determination of the stability of Cement Slurry under static Conditions.

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Decide the sequence for adding the additives to formulate mud.
CO2 - Determine the drilling fluid properties as per API standard
CO3 - Validate the experimental rheological properties with statistical models.
CO4 - Understand the interaction of drilling fluid with the subsurface formation
CO5 - Formulation of Cement Slurry.
CO6 - Determine the cement slurry properties for a given slurry.

TEXT / REFERENCE BOOKS

1. *API recommended practice 13B-2: Recommended practice for field testing oil-based drilling fluids* (8th ed.). American Petroleum Institute.
2. Bourgoyne, A. T., Jr., Chenevert, M. E., Millheim, K. K., & Young, F. S., Jr. (1991). *Applied drilling engineering*. Society of Petroleum Engineers.
3. Mitchell, R. F., & Miska, S. Z. (2011). *Fundamentals of drilling engineering*. Society of Petroleum Engineers.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**

24PE208T					Reservoir Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	-	-	100

COURSE OBJECTIVES

- To explain the basic concepts of reservoir media, fluid flow and its behaviour.
- To provide the difference between single and multiphase flow for reservoir flow system.
- To acquire and analyse the data of reservoir.
- To develop the reservoir performance evaluation.

Unit I: Rock Properties **Hours: 14**

Introduction to reservoir media–porous and fracture medium, concept of porosity fluid saturation, wettability, capillary pressure and relative permeability for understanding multiphase flow behavior in reservoir system, Salient features of Gas-Oil and Water-Oil relative permeability Curves, Factors affecting relative permeability, Three phase relative permeability, basic laboratory core data analysis for understanding petrophysical parameters.

Unit II: Flow Through Porous Media **Hours: 14**

Fundamentals of flow in porous media, Classification of flow system in porous media, Single phase and multiphase fluid flow in different state (steady and unsteady) and different systems (linear, radial, turbulent, spherical) considering compressible, slightly compressible and incompressible fluid, Diffusivity equation and its application for reservoir flow system

Unit III: Reservoir Drive Mechanism and Reserves Estimation **Hours: 12**

Reservoir Field operation data acquisition, Basic data of reservoir engineering; PVT data, Core data, Well logging and transient well testing information. Reservoir Drive Mechanism, Application of reservoir engineering principles: volumetric, material balance and decline curve analysis.

Unit IV: Introduction to Water Injection and Performance Analysis **Hours: 12**

Water injection concept, Fractional flow equation, Buckley Leverett equation, Pattern flooding, Reservoir performance analysis, Preparation of development schemes, IOR/EOR and workover jobs for reservoir.

Max <52Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Know the basics of fluid properties and petrophysical parameters of the petroleum reservoir system.
CO2 - Assess the importance of various petrophysical parameters for flow behaviour of fluids in multiphase petroleum reservoir system.
CO3 - Understand the different types of flow systems existing in a petroleum system in terms of rock, fluid and drive system.
CO4 - Analyse the importance of reservoir data acquisition activities in different stages of reservoir Analyse the importance of reservoir data acquisition activities in different stages of reservoir life
CO5 - Conceptualize sequential reservoir engineering activities.
CO6 - Design water injection strategy and analyse reservoir performance in conjunction with reservoir management concepts.

TEXT / REFERENCE BOOKS

1. Dake, L. P. (1978). *Fundamentals of reservoir engineering*. Elsevier.
2. Craft, B. C., Hawkins, M. F., & Terry, R. E. (1991). *Applied petroleum reservoir engineering* (2nd ed.). Prentice Hall.
3. Slider, H. C. (1983). *Worldwide practical petroleum reservoir engineering methods*. PennWell Publishing Company.
4. Ahmed, T., & McKinney, P. D. (2011). *Advanced reservoir engineering*. Gulf Professional Publishing.
5. Smith, C. R., Tracy, G. W., & Farrar, R. L. (1993). *Applied reservoir engineering* (Vols. 1–2). Oil & Gas Consultants International (OGCI) Publications.
6. Dandekar, A. Y. (2013). *Petroleum reservoir rock and fluid properties* (2nd ed.). CRC Press.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE209T					Well Log and Formation Evaluation					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	-	-	100

COURSE OBJECTIVES

- To provide the concept of various direct and indirect logging techniques.
- To identify geology and mineralogy of formations and determine its petrophysical properties.
- To estimate reservoir potential and production capacity and designing strategy for efficient exploitation.

Unit I: Introduction to Formation Evaluation and Well Logs **Hours: 10**

Introduction to petroleum formation evaluation methods; Introduction to well logs; LWD (Instruments and data recording); Well log tracks and their presentation; Types of cores, Routine and special core analysis techniques; Mud logging introduction; Description of cutting and oil shows in the log and Gas detection. Basic principle, data acquisition and interpretation of Temperature and Calliper Logs. Assignment on relevant topics.

Unit II: SP, GR and Resistivity Logs **Hours: 15**

SP log: Basic principle and log interpretation; GR lags: Basic principle, data acquisition and interpretation of gamma ray and spectral gamma ray logs; Resistivity Logs: tools, data acquisition and interpretation. Electrical image logs. Assignment on SP, GR, and Resistivity logs, Shale volume calculation and identification of gross sand and net sand, Rw calculation and its use in determining S_W.

Unit III: Porosity Logs- Density, Neutron, Sonic and NMR **Hours: 17**

Data acquisition and interpretation of Density porosity, Neutron porosity and Sonic logs; Porosity and lithology estimations (Neutron-Density, Sonic-Density, Sonic Neutron, MN and MID cross plots); Acoustic image logs and their applications; Nuclear Magnetic Resonance Log (Principles, Porosity and permeability estimation, type of fluid identification); Assignments to determine porosity, lithology and mineralogy from logs and their cross plots.

Unit IV: Summary of Interpretation After Integrating All Well Logs **Hours: 10**

Integration of different log data to determine lithology, shale volume, minerology, porosity, and net pay thickness; Evaluate reservoir potential and its production capability. Assignment or case studies on different formation types for evaluating the reservoir potential.

Max <52Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Describe the benefits and limitations of various direct and indirect formation evaluation techniques.
CO2 - Understand the basic principles, data acquisition, and interpretations of different well logging techniques.
CO3 - Distinguish the sub surface formation by predicting porosity, permeability and type of fluids from various well logs
CO4 - Estimate different reservoir properties (rocks and fluid properties) from individual logs.
CO5 - Integrate different log data and cross-plot to determine lithology, minerology, porosity and net pay thickness.
CO6 - Evaluate reservoir potential and its production capability for efficient exploitation.

TEXT / REFERENCE BOOKS

1. Rider, M. (2002). *The geological interpretation of well logs* (2nd ed.). Rider-French Consulting Ltd.
2. Serra, O., & Serra, L. (2004). *Well logging: Data acquisition and applications*. Serralog.
3. Jorden, J. R., & Campbell, F. L. (1986). *Well logging* (Vols. 1–2). Society of Petroleum Engineers.
4. Ellis, D. V., & Singer, J. M. (2007). *Well logging for earth scientists* (2nd ed.). Springer.
5. Darling, T. (2005). *Well logging and formation evaluation*. Gulf Professional Publishing.
6. Ellis, D. V., & Singer, J. M. (2007). *Well logging for earth scientists* (2nd ed.). Springer.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE202T					Heat and Mass Transfer					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	1	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand the different mode of heat transfer and its Industrial applications.
- To provide the importance of various modes of heat and mass transfer to solve various engineering problems.
- To explain the significance of mass transfer operations in various industries.
- To design the various mass transfer equipments.

Unit I: Conduction and Convection **Hours: 10**

Conduction: Steady-state and transient flow through various geometries, Convection: LMTD and NTU, overall heat transfer coefficient. Application of dimensional analysis to convection. Heat transfer rate and Heat transfer coefficient calculations. Double pipe parallel and counter flow heat exchangers, natural and forced convection through tubes and outside tubes, Shell and tube heat exchanger, and finned tube heat exchanger. Boiling of liquids and condensation of vapours.

Unit II: Radiation **Hours: 10**

Introduction, theories of radiation, electromagnetic spectrum, spectral emissive power, surface emission, total emissive power, emissivity; Radiative properties, emission, irradiation, absorptivity, reflectivity and transmissivity; Concept of black and grey body, radiation intensity, laws of black body radiation, non-black surfaces, radiation between black and grey surfaces.

Unit III: Diffusion **Hours: 10**

Diffusion in gases: Fick's law, determination and estimation of diffusion coefficient; diffusion through stagnant gas and equimolecular counter-diffusion. Diffusion in liquids: Mass transfer across phase boundaries, two-film theory and mass transfer coefficient.

Unit IV: Mass Transfer Operations **Hours: 09**

Gas Absorption, Adsorption, Extraction and Distillation (flash and differential): Basic principles, laws, and calculations. Equilibrium, co-current and counter-current operations. Ideal stage concept and calculation of number of ideal stages. Efficiency of Packed bed and tray columns.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand conduction and convection heat transfer equations to various geometry and solve complex engineering problems.
CO2 - Analyze the phenomena of boiling and condensation and determine the thermal coefficient of various heat exchanger configurations
CO3 - Comprehend and analyze the radiation heat transfer.
CO4 - Study the application of diffusional mass transfer in the reservoir field.
CO5 - Understand and design the various mass transfer operations in the oil and gas processing.
CO6 - Design the packed bed and tray column.

TEXT / REFERENCE BOOKS

1. Coulson, J. M., Richardson, J. F., Backhurst, J. R., & Harker, J. H. (2020). *Coulson and Richardson's chemical engineering: Volume 1, Fluid flow, heat transfer and mass transfer* (6th ed.). Elsevier.
2. Kern, D. Q. (2019). *Process heat transfer* (2nd ed.). Tata McGraw-Hill.
3. Holman, J. P. (2017). *Heat transfer* (10th ed.). Tata McGraw-Hill Education.
4. Çengel, Y. A. (2013). *Heat and mass transfer: A practical approach* (5th ed.). Tata McGraw-Hill Education.
5. Incropera, F. P., DeWitt, D. P., Bergman, T. L., & Lavine, A. S. (2007). *Fundamentals of heat and mass transfer* (6th ed.). John Wiley & Sons.
6. Yadav, R. (2004). *Heat and mass transfer* (5th ed.). Central Publishing House.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE210T					Earth Science and Hydrocarbon Exploration Field Work					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	6	3	-	-	-	-	50	50	100

COURSE OBJECTIVES

- To prepare students for the field of Petroleum Exploration
- To imply ability of reading maps, planning exploration and taking risk and decision
- To help students in identifying sedimentary rocks and its relation to petroleum system
- To able to help students in identifying and interpreting structural aspects in field.

LABORATORY COURSES

Practical classes/Field trips shall be based on theory course content of Earth science, Sedimentary geology, Petroleum Geology and Petroleum Exploration courses

Aim: Field familiarization of exploration in sedimentary basin and petroleum System

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Analyse, Identify and sample different minerals, rocks and fossils for detailed study
CO2 - Evaluate the structural aspects of an area
CO3 - Differentiate between Source, Reservoir and Trap rocks
CO4 - Perform the geological/Geophysical Mapping of a petroliferous basin
CO5 - Construct the geological and Geophysical maps of the area for exploration and exploitation
CO6 - Integrate and Evaluate the G&G data for predicting hydrocarbon resources.

TEXT / REFERENCE BOOKS

1. Coe, A. L. (2011). *Geological field techniques*. Wiley-Blackwell.
2. Compton, R. R. (1962). *Manual of field geology*. John Wiley & Sons.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**